A classification technique that is widely used is DNA base composition which is expressed as the percentage of Guanine plus Cytosine (G+C). It is a fixed property that reveals the degree of species relatedness. Ribosomal RNA sequencing is used to determine the diversity of organisms and the phylogenetic relationship. Basically ribosomes consists of two subunits, each of which is composed of protein and a type of RNA. Specific base sequences called as signature sequences are found in all groups of organisms. These unique DNA sequences are 5-10 bases long and found in 16s rRNA location and unique to major groups of prokaryotic organisms.

Nucleic acid based detection methods help in the detection of genomic materials. The 16s rRNA gene sequencing has been established as the "gold standard" for identification & taxonomic classification of microbial species.

8.5 The Three Domain System

This system of classification was introduced by C.Woese, O. Kandler and M.L.Wheelis, is an evolutionary model of phylogeny based on cells rRNA sequences(differences in the sequences of nucleotides) studies. They grouped all living organisms into three domains: Bacteria, Archaea and Eukaryota (Figure 8.5).

Bacteria and Archaea are two different groups of prokaryotes. The domain Bacteria comprise the vast majority of prokaryotes. The domain Archaea contains prokaryotes that live mostly in extreme environments. The domain Eukaryota contains living organisms that includes Kingdom Protista, Kingdom Fungi, Kingdom Plantae and Kingdom

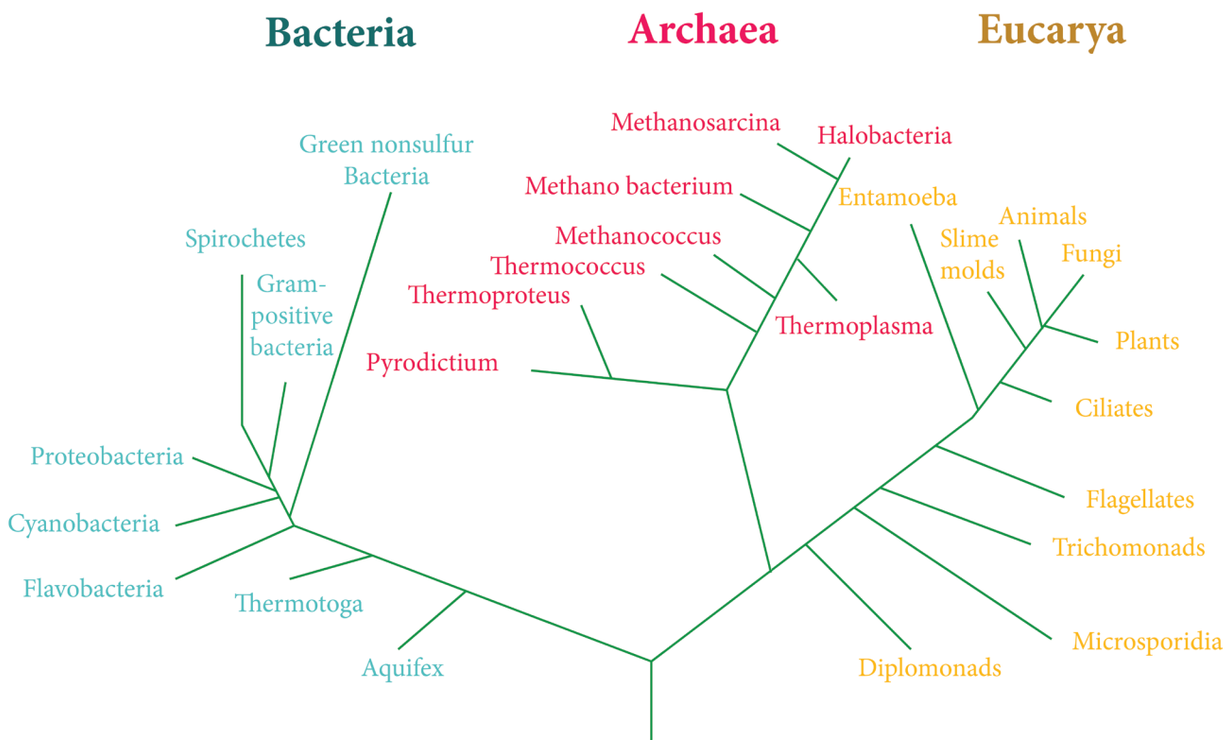


Figure 8.5: A phylogenetic tree based on rRNA analysis. Organisms are classified into three domains: Bacteria, Archaea and Eukaryotes as proposed by Carl Woese et al.

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